

When a Pandemic Comes to Town: Combating Propagation of Novel Viruses in Russian Cities Using Statistical and Mathematical Modeling

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Abstract—In a contemporary world with its never-ending threat of emergence of novel viruses with a pandemic potential, epidemic surveillance is an essential component of modern healthcare. To better fulfill its duties, epidemic surveillance should rely on quantitative approaches. The problem is that there exists a certain gap between the theoretical modeling approaches, the recommendations of which might not be applicable in real life due to many reasons, and the long-established practical methods of monitoring and controlling disease propagation which in some cases happen to be ineffective. In this paper, the author proposes a concept of a modeling framework built on top of existing fragmented modeling solutions, which helps unify and quantify intertwined processes of disease control on different levels. The framework is intended to be parsimonious and data-driven, which will facilitate its usage within Russian epidemic surveillance system. The paper demonstrates how to connect different layers of the outbreak response within the framework and shows the methods of its city-level usage in the context of emerging acute respiratory infections with pandemic potential, such as all variants of severe acute respiratory syndrome, including COVID-19, and pandemic influenza.

Keywords—mathematical epidemiology, COVID-19, influenza, agent-based modeling, statistical modeling

I. INTRODUCTION

In a contemporary world with its never-ending threat of emergence of novel viruses with a pandemic potential, epidemic surveillance is an essential component of modern healthcare. According to [1], two main goals of surveillance is to provide early warning about a potential pandemic, e.g. by global monitoring and rapid identification of domestic introductions, and to provide support for decision making during an ongoing pandemic, including tracking incidence, prevalence, and the pathogen's properties. "The first set of decisions faced by a domestic public health jurisdiction, following the appearance of a pandemic threat somewhere else in the world, concerns the questions of whether, how, and to what extent to scale up a response to reduce the risk of importation or, if importation has happened, to control its spread within the jurisdiction. Measures to reduce importation via restriction or testing of inbound travelers may buy limited time to prepare, though such measures lose relevance once local transmission is established. To inform decisions about how to balance scarce public health resources

between preventing importation vs. controlling local spread, it is critical to assess the risk that the infection has already arrived and started spreading within the jurisdiction" [1].

To better fulfill its duties, epidemic surveillance should rely on quantitative approaches. The problem is that there exists a certain gap between the theoretical modeling approaches, the recommendations of which might not be applicable in real life due to many reasons (among the most popular the lack of information and resources), and the long-established practical methods of monitoring and controlling disease propagation which in some cases happen to be ineffective. A typical example is syndromic and questionnaire-based screening programs for travelers, which are still in use despite their ineffectiveness according to empirical evidence [2].

In this paper, the author proposes a concept of a modeling framework built on top of existing fragmented modeling solutions, based on his experience with developing models for COVID-19 surveillance in St Petersburg in 2020-2021. The proposed framework serves an aim of unifying and quantifying intertwined processes of disease control on different levels. The framework is constructed in a way to be parsimonious and data-driven, which will facilitate its usage within Russian epidemic surveillance system. The paper demonstrates how to connect different layers of the outbreak response within the framework and shows the methods of its city-level usage in the context of emerging acute respiratory infections with pandemic potential, such as all variants of SARS (severe acute respiratory syndrome), including COVID-19, and pandemic influenza.

II. REVIEW

Due to the extreme urgency of the problem of planning effective comprehensive measures to combat pandemics, many research groups around the world work on this problem. In terms of research conducted in the 21st century, the first major event that motivated a comprehensive quantitative and qualitative assessment of containment measures was the threat of terrorist biological attacks in the United States after the September 11 terrorist attack, associated with the potential spread of anthrax and other infectious agents. In this regard, in the mid-2000s, health authorities refined recommendations for actions in such situations and developed many comprehensive response systems based on mathematical and computer models (see [3], [4]). Other

challenges that have stimulated the optimization of epidemic containment planning systems based on quantitative methods include the outbreaks of atypical pneumonia caused by the SARS-CoV (2002-2003), Ebola (2013-2016), Middle East respiratory syndrome (2015), and the A(H1N1)pdm09 influenza pandemic (2009). Since the beginning of 2020 and up to the present day, a large number of studies have been published on the dynamics and control of COVID-19. The following three methodologies are commonly used as a quantitative research apparatus used to systematize the action plan of health authorities for epidemic outbreaks:

- Statistical approaches to risk assessments (an old but still relevant qualitative review of these methods can be found in [5]).
- Metapopulation modeling of the spread of infections between countries and regions due to population migration. For example, the use of the GLEAM model of the MOBS-Lab laboratory from Northeastern University, USA [6] to predict influenza and COVID-19 pandemics
- Multiagent modeling of epidemic outbreaks in particular jurisdictions (city-level, district-level) and in limited territories (for example, cruise ships exposed to infection, as well as medical institutions treating COVID-infected people), which precede the transition of infection to the main population (e.g., [7])

In connection with the development of artificial intelligence technologies, real-time assessment of the effectiveness of anti-epidemic measures based on machine learning has gained popularity [8]). This approach has been applied to forecasting COVID-19, for example, see [9].

A promising approach that can potentially detect outbreaks of new infections when processing statistical epidemic surveillance data is the approach based on the theory of critical events, which, despite its novelty and fairly high degree of generality, demonstrates practical significance as a basis for early response systems to epidemic threats (see [10], [11]).

It is important to note that the conclusions about the effectiveness of certain methods of monitoring and preventing pandemics may differ significantly depending on the methods used for analysis (see, for example, [12], [13]) even when the same data are used in the models. Different types of modeling and analysis methods used can produce different incidence predictions (and, as a result, different estimates of the effectiveness of containment measures), as demonstrated, for example, by comparative tests of different predictive models of influenza dynamics, regularly conducted by the Centers for Disease Control in the United States [14]. Conducting research on modeling and containment of pandemics by a large number of different research teams helps to minimize the error associated with deviations in data and the peculiarities of the methods. In the case of risk analysis and pandemic forecasting in the Russian Federation, this is even more relevant, since the number of quantitative analysis works based on Russian data is currently very limited.

III. METHODS

The modeling framework is implemented as a chain of three interconnected models, its structure is presented in Fig.

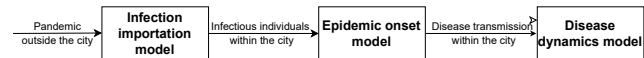


Fig. 1. Framework scheme.

1 and Table I. The included submodels cover the whole process of an epidemic progression in a city, starting from the moment of establishing the fact of ongoing pandemic somewhere in the world and finishing by the fact of stable disease transmission within the regarded city. The author omits the case when the regarded city is a source of a novel pandemic (e.g., like a case of Wuhan in 2019-2020) due to the fact that such events are rare and moreover not very suitable for modeling due to lack of clear quantitative information. The models within the framework are intended to subsequently answer the following questions:

- What is a risk of infection introduction into the city, depending on the epidemic situation in the adjacent regions, the intensity of travel flows and introduced control measures?
- What is a risk of a full-grown epidemic outbreak in the city, depending on the virus properties, current registered number of infections within city limits and introduced control measures for detection and isolation of the infectious?
- What is the prospected dynamics of an outbreak after its inception, depending on the virus properties, contact patterns in the population and introduced control measures to limit the disease propagation?

The resulting system is capable of performing quantitative analysis of the effectiveness of containment and control measures at the city level and can be used for decision support.

A. Infection Importation Model

The model of infection importation is based on the Baroyan-Rvachev approach [15], [16], where the infection transmission between the communicating geographical regions is set with the help of metapopulation modeling. A modern GLEAM model [6], earlier mentioned in 'Review' section, is built according to the same principle. Knowing a number of infected people in the regions already affected by a pandemic and migration flows between the cities, using which one can set a correspondence matrix in the model, we can infer the distribution of random variables which define the time until arrival of first infected individuals and subsequently the intervals between those events, which gives us a picture of disease introduction attempts. The author assume the usage of the model implemented in [17]. In the cited work it was established that the data on inter-city commute available to date were not very accurate, as a result, the correspondence matrix with exact values of migration could be replaced by a much simpler gravity model of migration where it is assumed that the interaction of the populations is proportional to their product and inversely proportional to the square of the distance between them.

B. Epidemic Onset Model

In scientific literature, it is often the case when an epidemic onset and subsequent disease dynamics are simulated using a single model, i.e. and epidemic onset model and a disease dynamics model are not separated. However,

TABLE I. FRAMEWORK DESCRIPTION

Model	Aim	Methods
Infection importation model	Assess the probability of infection introduction into the city, simulate measures to monitor and prevent introduction	Statistical approximation of the probabilities of isolation of potential first infected subjects at the moment of their arrival through the transport infrastructure. If necessary, multi-agent models of transport infrastructure (airports/train stations) can be used in a more detailed version to plan isolation measures)
Epidemic onset model	Timely detect the disease inception before the formation of stable chains of infection transmission and calculate economically feasible targeted (tracking contacts of patient zero when detected) or generalized (isolation of the population as a whole) measures to break these chains;	A multiagent model or a statistical model based on the theory of critical phenomena to assess the probability of the infection transitioning to the active phase (the formation of stable chains of infection transmission).
Disease dynamics model	In the event of further spread of the disease, implement the most economically and socially justified measures to reduce damage from the infection, including those related to quarantine and vaccination (if vaccines are available).	A multi-agent model for the initial phase of infection (taking into account the heterogeneity of the population by the average number of contacts, i.e. the presence of risk groups and so-called superspreaders, who are distinguished by abnormally high social activity).

considering the application scenario of the framework described in this study, the author considers it crucial to differentiate those two processes. The reason is that the models which might satisfactorily reproduce the epidemic outbreaks as a whole often fail to accurately describe its first phase and thus might confuse the decision makers. For instance, in Fig. 2, where a trajectory of a standard SIR model is fitted to observed COVID-19 incidence data of the first epidemic wave in St Petersburg, one can see that during several initial weeks the course of disease dynamics cannot be explained well by a model due to the stochastic nature of an epidemic onset. As a result, it is important to use stochastic approaches for an epidemic onset model, while this is not necessary to describe a massive epidemic outbreak.

1) *Multiagent models:* As an example of simple methods to construct an epidemic onset model, two approaches presented in [18] might be mentioned. To test traveler screening effectiveness, the researchers considered how many additional infected travel attempts could be tolerated before causing an outbreak with high probability in screening vs no-screening scenarios (ΔN). To calculate ΔN , the researchers used theory from stochastic processes about the long-term probability of extinction to compute the number of infected travelers required to cause an outbreak with a fixed probability p . Thus, this approach might be used to find a distribution of an absolute value N , i.e.,

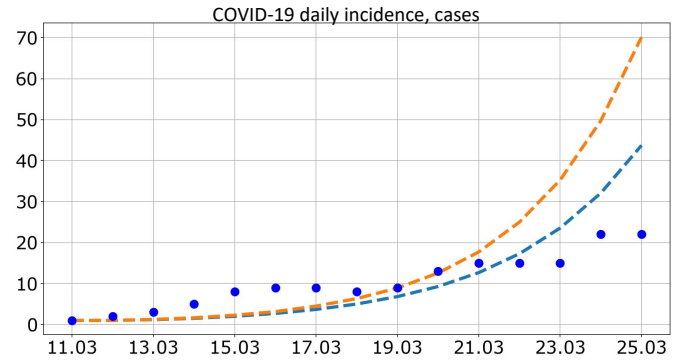


Fig. 2. Incidence cases of COVID-19 during the epidemic onset in St Petersburg, 2020. Daily incidence is shown by blue dots, orange and blue dashed lines show the modeling assessment made by a SIR-type differential mode.

number of infected travel attempts before the outbreak onset, depending on the selection of the input parameters. In the second approach, the authors calculated how much longer it takes for an outbreak of size X to occur at the destination in screening vs no-screening scenarios (Δt). To calculate Δt , they assumed that the number of infected travelers arriving at the airport followed a Poisson process, with mean λ infected travelers per day. The transmission chains initialized by infected travelers were simulated at the destination until X infections had occurred. One might use this approach via inferring distributions of t .

In our framework, we assume the usage of a multiagent model which simulates the simultaneous propagation of several respiratory viruses in a population [19]. The ability of simulate multistrain infections is useful when we consider the arrival of a novel virus which cannot be distinguished by the caused symptoms from the already existing diseases and thus might be 'hiding' in the general pool of infections. Particularly, this is an important case for pandemic influenza and SARS-CoV.

To serve as a proof of concept, a set of simulations was performed with the model to assess the role of different factors in the surge of the outbreaks in the population from the imported virus strain. We assumed that 3 different disease strains are circulating in the population, denoted as $m = 1, 2, 3$. A synthetic population of St Petersburg is used in the simulation to accurately represent the contact patterns typical for the city. To calculate the number of occurred outbreaks, we relied on a value of disease prevalence at day 15 — which roughly corresponds to two weeks after the virus introduction. It is considered that disease trajectories which show zero prevalence at day 15, i.e. $I(15) = 0$, mean the absence of the outbreak (the stable transmission chain was not established, and the disease died out).

a) Dependence on initial number of infected:

Expanding our earlier research, we quantified the role of the initial number of the infected people introduced in the population on the subsequent rise of disease prevalence. In our experiment, we varied the number of initially infected for one strain: (a) $I_0^{(m)} = 1$, (b) $I_0^{(m)} = 10$, (c) $I_0^{(m)} = 100$, where $m \in \{1, 2, 3\}$ is the index of the virus strain. Thus, the overall number of initially infected was 3, 30 and 300 correspondingly. The research question was: to what extent the number of the established outbreaks (i.e. cases when

$I(15) > 0$) depends on I_0 . The experiment results are shown in Fig. 3.

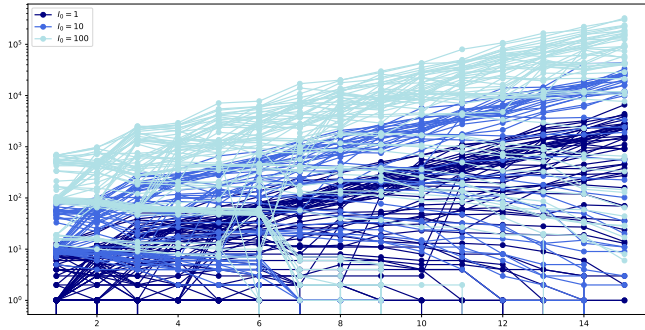


Fig. 3. Simulated disease prevalence during the disease onset (days 1-15) depending on the number I_0 of initially infected individuals (logarithmic scale)

As it can be seen, the typical types of prevalence trajectories include those which show the initial stage of an exponential growth and those which are on the decline due to breaks in infection transmission chains. As a result, the epidemics started from a single 'patient zero' of each strain has higher chances of dying out, and the prevalence levels are largely defined by an initial value of I_0 which is evident by looking at the stripes of the same color in the figure. This visual analysis is also supported by the numerical assessment. It is known that in SEIR-type deterministic models the introduction of infection is often modeled by one person, i.e. $I_0 = 1$, since in these types of models the initial surge of disease prevalence caused by a single individual is guaranteed, if the selected parameter values allow it [20]. At the same time, in the stochastic models, starting the simulation from one 'patient zero' might lead to the infection dying out rapidly. In our case, for the population with 5% of susceptible individuals, half of the started outbreaks die out till day 15 [19]. The next planned step in this research is to analyze the result more thoroughly by iterating through I_0 with smaller step and trying different network topologies.

b) Dependence on choice of the initially infected:

By default, the initially infected people in the model are distributed independently within the population for each simulation run, which seem to conform to the general principles of infection introduction in the population (each time random people become 'patients zero'). In this set of simulations, we have made a set of initially infected people equal to all the simulation runs in order to assess the impact of this factor on the subsequent disease progression. In Fig. 4 the disease trajectories of the simulation runs starting from the same individuals compared to the conventional simulation run trajectories.

Somewhat unexpectedly, in our simulations the randomized distribution of initially infected individuals does not seem to add up anything to the uncertainty of the outcome related to the epidemic dynamics. The usage of the randomized and the fixed I_0 distribution results in almost indistinguishable means (7405 and 7779.8) and standard deviations (11301 and 11670.6) for the disease prevalence at day 15. Thus, the uncertainty factor connected with the distribution of the initially infected individuals does not seem to play a crucial role in the simulated epidemics, which

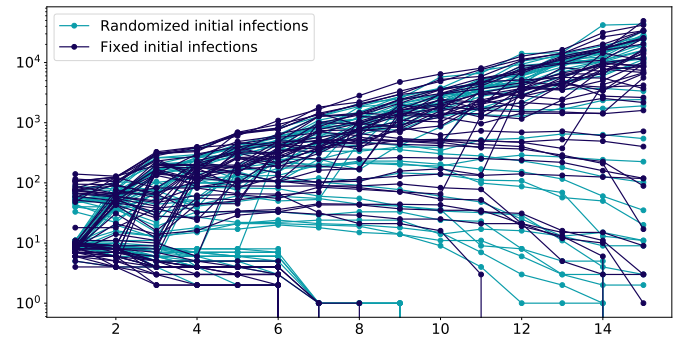


Fig. 4. Simulated disease prevalence during the disease onset (days 1-15) depending on the distribution of initially infected individuals (logarithmic scale)

somewhat contradicts the idea of superspreading events (the more connections a 'patient zero' has — the higher is the chance that this individual will start an epidemic outbreak). It is important to notice that in the simulation we considered that only 5% of the population is susceptible to the virus, which leads to clustered contact network (a lot of links become not significant in terms of disease propagation, because one or both nodes are not susceptible). This fact must cause large non-uniformity in the degree distribution, which means that one randomly selected individual is very different from the other in terms of his ability to become a superspreader. Still, it seems that the effect is not powerful enough when the susceptible population size is $\sim 10^4$.

2) *Simpler Models*: It is not always possible to use multiagent models, especially in cases of absence of enough data to calibrate it. As an example of a simpler method to construct an epidemic onset model, the author wants to refer to two approaches which are presented in [18]. To test traveler screening effectiveness, the researchers considered how many additional infected travel attempts could be tolerated before causing an outbreak, with high probability in screening vs no-screening scenarios (ΔN). To calculate ΔN , the researchers used theory from stochastic processes about the long-term probability of extinction to compute the number of infected travelers required to cause an outbreak with a fixed probability p . Thus, this approach might be used to find a distribution of an absolute value N , i.e., number of infected travel attempts before the outbreak onset, depending on the selection of the input parameters. In the second approach, the authors calculated how much longer it takes for an outbreak of size X to occur at the destination in screening vs no-screening scenarios (Δt). To calculate Δt , they assumed that the number of infected travelers arriving at the airport followed a Poisson process with mean λ infected travelers per day. The transmission chains initialized by infected travelers were simulated at the destination until X infections had occurred. One might use this approach via inferring distributions of t .

For our framework, we have at our disposal an already implemented simplified stochastic compartmental model used for studying tuberculosis propagation caused by discrete influxes of infectious persons into a certain region [21]. Within the simulation, it is possible to assign a number of people in a group which comes to the city and an interval between the events of their arrival. The model

allows assessing the thresholds for the parameter values, which divide cases of a successful epidemic onset from the cases when the outbreaks do not occur. For sufficiently rare immigration inflows the particular portion of afflicted persons in the group of individuals coming to the region does not influence the dynamics of the infected subgroups. Varying the values of the parameters $g_H(t)$, the immigration inflows of individuals, and the mean period of periodic migration flows m_v , one can estimate the admissible mean level of immigration inflows of infectious individuals which won't cause the incidence growth. This method might be useful to plan quarantine measures for the incoming travelers.

C. Disease Dynamics Model

One of the main drawbacks of the multi-agent approach which seriously limits its application is related to excessive demand of computational power to handle the experiments with the model. Even in the simplest case, when the aim is to calculate the disease trajectory for one outbreak at the city scale with a predefined set of parameter values, several simulation runs are required to address stochastic uncertainty. As a result, the experiment may last from hours to days, depending on the model employed and the computational resources available. At the same time, when we talk about an ongoing disease outbreak with massive number of infected, the stochasticity does not play such an important role, as it is in the case of modeling a new epidemic onset. In fact, when the number of newly infected is large, the difference in the outputs of the detailed multi-agent models and much simpler compartmental ones is often negligible [22] which makes it possible to locally replace the stochastic models by the classical SIR-type deterministic models without loss of disease dynamic reproduction accuracy. To perform a matching of these two distinct approaches without negatively affecting the accuracy of simulation, a hybrid modeling approach is recommended [23], [24].

In our framework, we use a hybrid model implemented by the author for modeling the circulation of ARI¹. If the computational resources available make it possible to perform multiagent simulation for large numbers of infected persons in the population, the model may continue to operate in a multiagent mode until the prospected end of the outbreak. Otherwise, the switch to compartmental modeling will occur which increases the simulation speed.

IV. CONCLUSION

In this paper, the author presented a modeling system for quantitative analysis of the effectiveness of measures to contain and control infections, implemented as a chain of interconnected models. The significance of the results for the development of this framework lies in demonstrating the effective practical application of earlier developed approaches (assessment of the uncertainty of forecasts and the inaccuracy of estimates of epidemic indicators), as well as obtaining important conclusions on the specifics of

using quantitative systems for describing the dynamics of epidemic outbreaks of new types of viruses in the practice of real epidemic surveillance, which will ultimately improve the effectiveness of preventing and containing epidemics, and verify model assumptions about the characteristics of the multiphase process of spreading infectious agents under the introduction of control measures of containment. The author's aim was to describe the framework in general as a conglomerate of interacting models with keeping in mind its flexibility which allows replacing submodels when we see fit. Due to that reason, the text deliberately lacks detailed description of the submodels. In the future we assume to allow the selection of submodels dependent on available input data using the indicators of optimal structural complexity.

The limitation of the current version of the framework is that it does not allow interval assessment of the output values to measure prediction uncertainty. Since the techniques of uncertainty assessment were already developed for the submodels used in the framework, we do not consider it a different task to add interval assessment in the nearest future.

Also, at the moment, we do not consider explicitly the detection of infectious individuals in the models, which might not be very important when using abstract synthesized data, but is very crucial for the planned model calibration to the retrospective data of real outbreaks. According to [18], the screening effectiveness of travelers dramatically affects the epidemic situation, and, moreover, it is highly variable for different pathogens (e.g., SARS-CoV-1, Ebola, SARS-CoV-2, and influenza).

The results presented in the paper are a part of a project which is aimed at a comprehensive analysis of the dynamics of epidemic morbidity in St. Petersburg, taking into account the impact of population heterogeneity. The implementation of the project will allow systematizing the set of measures taken to control morbidity at the level of qualitative and quantitative description, with a choice of the possible level of detail of the description depending on the availability and quality of the available data, as well as assessing the quality of the collected data and formulating recommendations for their improvement. The availability of a detailed qualitative description of the dynamics of the epidemic process confirmed by data will provide a tool for working out various scenarios for the course of the epidemic depending on the restrictive measures taken in real time. Based on a reliable quantitative description, it becomes possible to justify the use of statistical estimates and mathematical models of varying degrees of detail, which will allow calculating quantitative indicators associated with the risks of a pandemic and the dynamics of its spread when this risk is realized, and, as a consequence, a quantitative calculation of the effectiveness of containment measures.

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